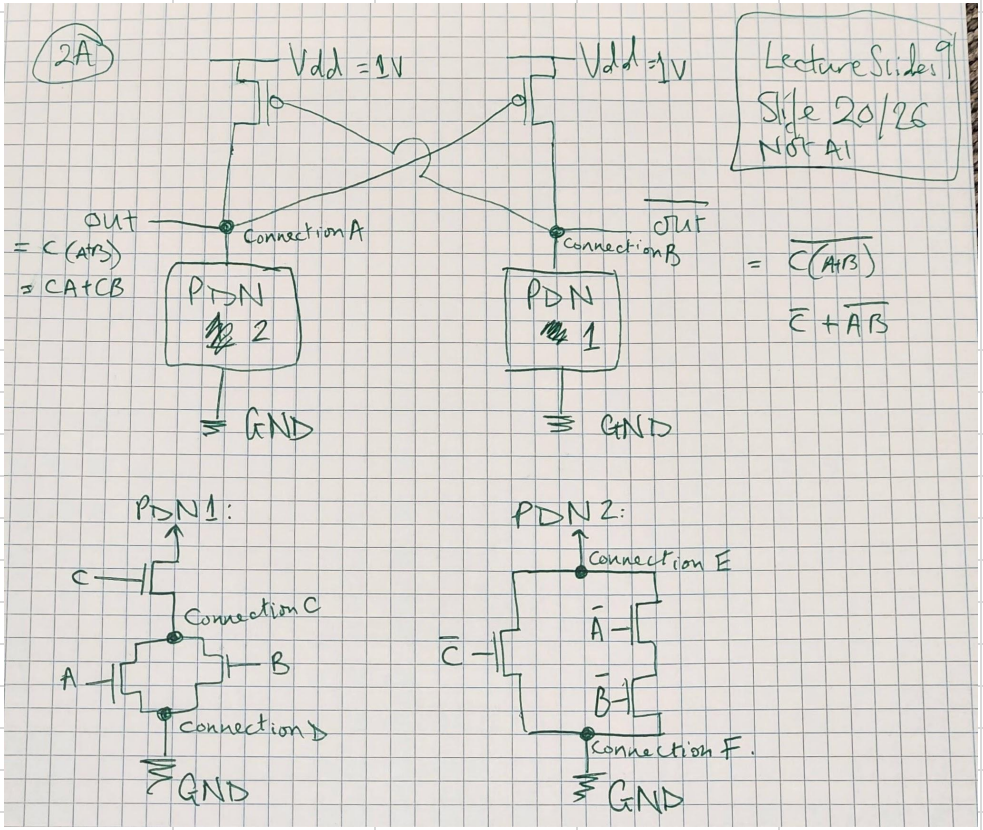
2A

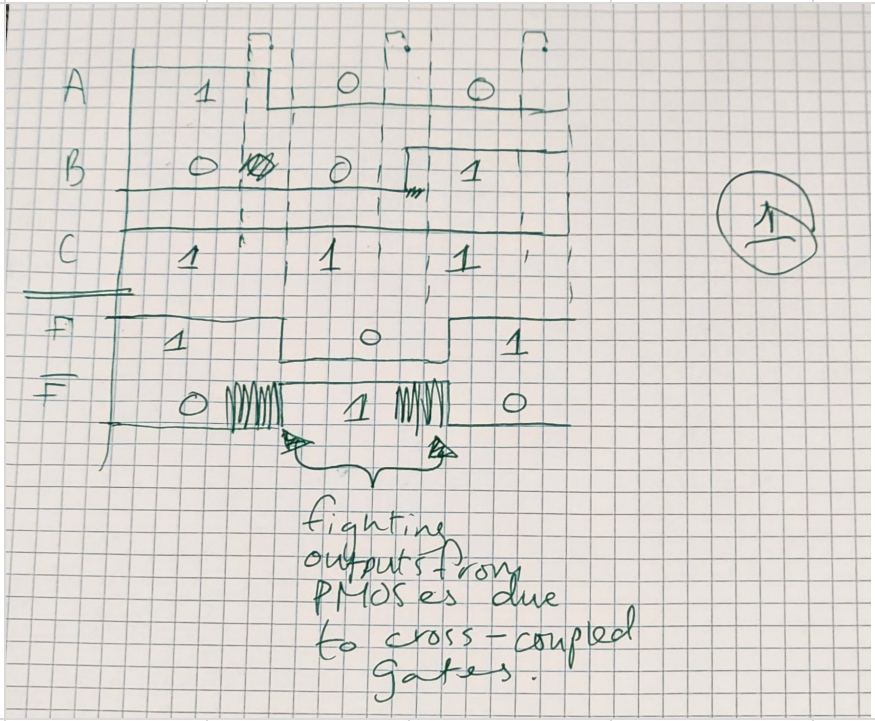


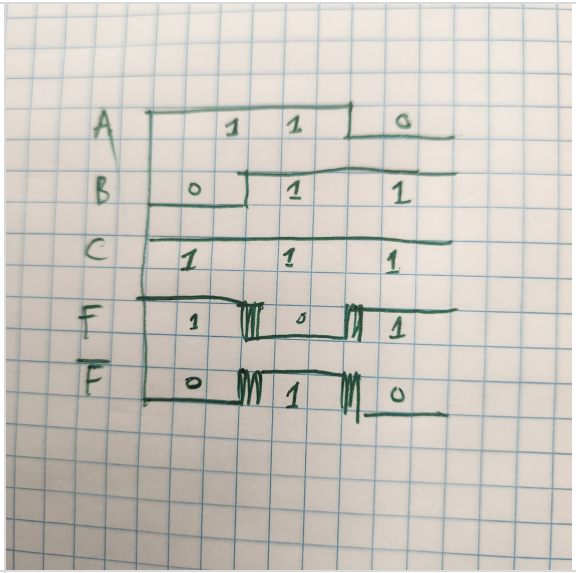

AI used for comprehension of topic here, not answer of problem (as suggested by Prof Joshi). I am not confident this answer is correct at all.

F=CA+CB

| | |
|---|---|
| ABC = 101 | ABC = 001 |
| PDN1 conducts, PDN2 does not. | PDN2 conducts, PDN1 does not. |
| PDN1 links to ground, PDN2 does not. | PDN2 connected to ground makes F 0. |
| F = 1, Fbar = 0 | Fbar = 1 |

Both outputs can't stably rest at 1, because the two PDNs are strictly complementary. If the output is one, then that means that the PDN associated equals zero, which means that the other PDN would pass current through, which would drain the voltage from the alternate current. The two PMOSes make a vicious negative cycle.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2B | Scenario I: A : 1 →0 occurs before B : 0 →1. | | | | | | |
| | (A,B,C) = (1,0,1) to (0,1,1), with C = 1 throughout. | | | | | | |
| | States: | 101 | "001" | "011" | | | |
| |  | | | | | | |
| | | | | | | | |
| | Scenario 2: B: 0 →1 occurs before A : 1 →0. | | | | | | |
| | (A,B,C) = (1,0,1) to (0,1,1), with C = 1 throughout. | | | | | | |
| | States: | 101 | 111 | "011" | | | |




The uncertainties here also stem from the fighting cross-coupled PMOSes.

To solve this issue, the circuit could introduce a clock? I'm not sure how sensible an idea this is though. It would lock values to pause propagation which would cause values to be sampled not pingponged.